

Explainability Analysis

of a Chess Position Classification CNN

Counterfactual Explanations and Gradient Saliency

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Main idea

The CNN is not final goal. It is the model that I explain.

- The model classifies chess positions into three evaluation classes.
- Accuracy is used only to check that model learned signal.
- Main focus is explanation: what does model react to, and how stable are decisions?
- I use saliency as baseline and counterfactuals as main XAI method.



Main XAI Question

What small board change is enough to change model prediction?

This is not trying to build strongest chess engine. It is about model behavior.

- ① Can small CNN trained on chess tensors learn enough structure to be useful for explanation?
- ② Do gradient saliency and counterfactual explanations show different parts of model behavior?
- ③ How stable are model predictions when small piece-removal changes are made?

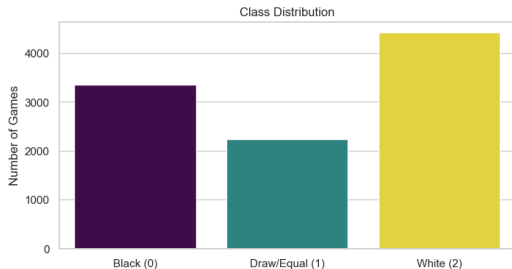
Important point

Explainability is not only making heatmap. Useful explanation should also test whether changing input actually changes model decision.

- Dataset: **10,000 Lichess positions**
- One position extracted after **20 full moves**
- Input: $13 \times 8 \times 8$ tensor
- 12 piece channels + 1 side-to-move channel
- Labels made with Stockfish evaluation

Label rule

- Black better: $\text{eval} < -1.0$
- Draw/Equal: $-1.0 \leq \text{eval} \leq +1.0$
- White better: $\text{eval} > +1.0$



Class distribution is not perfectly balanced, but all classes are represented.

Pipeline

- 1 Parse PGN games.
- 2 Extract position after 20 full moves.
- 3 Evaluate position with Stockfish.
- 4 Convert board into tensor.
- 5 Train CNN classifier.

XAI Analysis

- 1 Compute gradient saliency map.
- 2 Remove one non-king piece.
- 3 Check if prediction changes.
- 4 Measure material distance.
- 5 Compare stability across classes.

Accuracy

model check

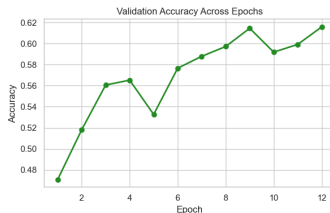
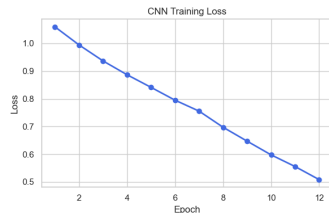
Distance

counterfactual size

Stability

decision fragility

CNN Training Behavior



- Training loss decreases steadily.
- Validation accuracy improves over epochs.
- Final validation accuracy is around **61.60%**.
- This shows model learned signal from board tensors.

Interpretation

The CNN is not perfect evaluator, but it is strong enough to be explained.

Model Performance on Test Set

63.73%

Test accuracy

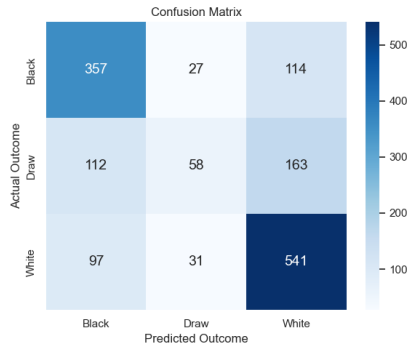
44.60%

Majority baseline

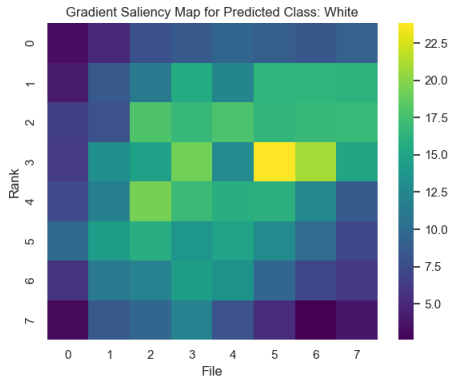
+19.13

Percentage points

The model performs clearly above baseline. Draw/Equal is hardest class because it is between Black-better and White-better positions.



Baseline Explanation: Gradient Saliency



What it shows

- Local sensitivity of predicted class score.
- Gradients are summed over 13 channels.
- Brighter square means larger local effect.

Limitation

Saliency does not prove that changing square will flip prediction.

This is why counterfactual test is needed next.

Counterfactual question

What minimal change to input would make model predict different class?

Implemented method

- Level 1 counterfactuals.
- Remove one non-king piece.
- Check if prediction flips.
- Measure material distance.

Why this is useful

- Simple and interpretable.
- Directly linked to chess material.
- Shows model boundary.
- Exposes fragile decisions.

Removing piece is not legal chess move, but here it is sensitivity test for model.

76.20%

flip rate on 500 test positions

Positive side

- Method finds many valid counterfactuals.
- One-piece interventions often change prediction.
- This gives many examples to analyze.

Important caution

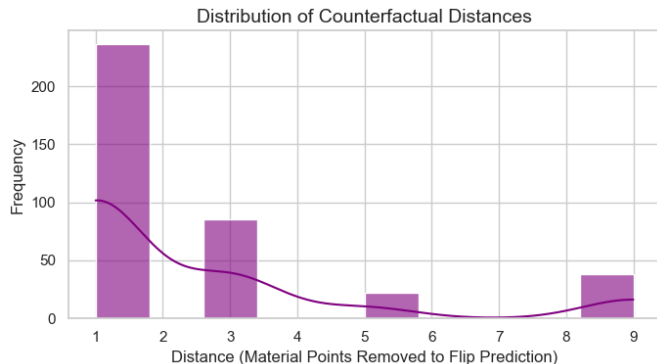
- Very high flip rate also means fragility.
- Model changes decision too easily.
- This is not how strong chess engine should behave.

Piece	Count
White Pawn	144
Black Pawn	92
White Bishop	32
White Queen	25
White Knight	22

- Pawns appear most often.
- Low material changes frequently flip model.
- Queen removals also appear, so not all flips are small.
- This reveals model sensitivity, not real chess truth.

Key point: high counterfactual validity also exposes model weakness.

Counterfactual Distance Distribution

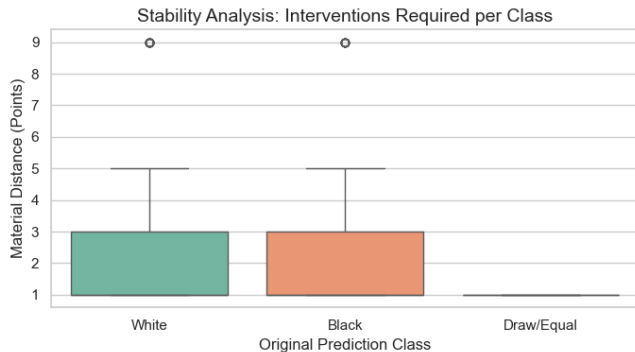


- Most distances are 1 or 3.
- Many flips come from pawn or minor-piece removal.
- Some flips require 5 or 9 material points.
- Low-distance flips show boundary is often close.

Interpretation

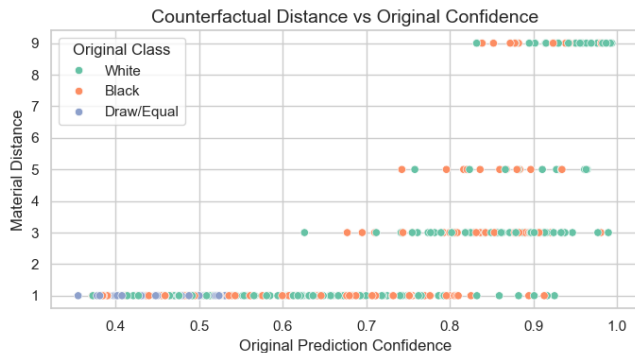
This supports fragility of CNN decisions under small board changes.

Stability by Predicted Class



Average distance

- Black: about **2.46**
 - Draw/Equal: about **1.00**
 - White: about **2.81**
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- All averages are low.
 - Draw/Equal flips with smallest distance.
 - White and Black produce more valid counterfactuals.



- Low-distance counterfactuals appear across many confidence levels.
- Some confident predictions still flip after small change.
- Confidence alone does not guarantee stable decision.

XAI finding

The model can be confident and still fragile.

Component	Main Result	Interpretation
Dataset	10,000 positions after 20 full moves	Realistic XAI experiment
Model	63.73% accuracy vs 44.60% baseline	Meaningful but not perfect
Saliency	Gradient heatmap	Local sensitivity
Counterfactuals	76.20% flip rate	Valid examples, but fragile model
Distance	Mostly 1 or 3 points	Low-material sensitivity
Stability	Low average distances	Not highly stable

Main finding: counterfactuals show both model behavior and model weakness.

Method limitations

- Piece removal is not legal chess move.
- Material distance does not fully measure positional importance.
- Pawn near promotion can matter more than material value.

Model limitations

- CNN is not strong chess evaluator.
- Draw/Equal class is difficult.
- High flip rate shows fragile boundaries.

- CNN learned meaningful chess evaluation structure.
- Saliency shows local board sensitivity.
- Counterfactuals test concrete piece-level interventions.
- High flip rate gives many valid examples, but also shows fragility.
- Counterfactual explanations are useful because they show where model is unreliable.

Thank you